
CE 302: Geotechnical Engineering

Course Code	CE 302
Course Name	Geotechnical Engineering
L-T-P-C	3-0-2-4
Prerequisites	None
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)
Recorded to	CE 311 by the 52nd BoA (Nov 2023)

Course Contents

- **Stress Distribution in Soils**

Effective stress concept, water table, seepage and capillarity effects and failure of structures by piping. Boussinesq theory-pressure distribution diagram on a horizontal and vertical plane, pressure bulb. [5 Lectures]

- **Shear Strength of Soil**

Stress strain curve, stress at a point-Mohr circle of stress, Mohr-coulomb failure criteria, pore pressure, total and effective stress. Peak and residual shear strength. Factors affecting shear strength, Laboratory measurement of shear strength by box shear, unconfined, Vane shear test and triaxial tests under different drainage conditions. Shear strength characteristics of sands and clays. [7 Lectures]

- **Subsoil Exploration**

Methods of subsoil exploration Direct, semi direct and indirect methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of Boring, Types of samples, Criteria for undisturbed samples, Transport and preservation of samples, Borelogs, planning of exploration programmes, report writing. [6 Lectures]

- **Earth Pressure**

Types of Earth pressure. Rankines and Coulomb's Active and passive earth pressure in soils. [4 Lectures]

- **Module V**

Bearing Capacity Safe bearing capacity and allowable bearing pressure, General and local shear failure conditions, Terzaghi's bearing capacity equations its modifications for square, rectangular and circular foundation, Factors affecting bearing capacity of Soil.

Shallow Foundations: Factors effecting locations of foundation and design considerations of shallow foundations, choice of type of foundations, Foundations on expansive soils. [9 Lectures]

- **Settlement analysis**

Causes of settlement, Computation of settlement, allowable settlement. Measures to reduce settlement. [3 Lectures]

- **Pile Foundations**

Types, Construction, load carrying capacity of single pile – Dynamic Formula, Static formula, Pile load tests, Load carrying capacity of pile groups, settlement of pile groups, Negative skin friction. [8 Lectures]

Laboratory Modules

1. Determination of Specific Gravity
2. Grain Size Analysis
 - A. Sieve Analysis
 - B. Hydrometer Analysis
3. Determination of Consistency Limits
4. Permeability Test
 - A. Constant Head Method
 - B. Falling Head Method
5. Proctor Light And Heavy Compaction Test
6. Direct Shear Test
7. Unconfined Compression Test
8. Unconsolidated and Undrained Triaxial Test
9. Vane Shear Test
10. CBR Test
11. Field Density Test
12. Free swell and consolidation test

Text Books

1. Shamsheer Prakash, Gopal Ranjan and Swami Saran, **Analysis, Design of foundations and Earth retaining structures**, IBH Publishers, 1979.
2. Braja M. Das, **Principles of Foundation Engineering**, Cengage Learning, 2015.
3. V.N.S. Murty, **Geotechnical Engineering**, Book World Enterprises, 2002.

References

1. W.C Teng, **Foundation Design**, Prentice Hall Publishers, 1962.
2. J. E. Bowles, **Foundation Analysis and Design**, TataMc-Graw Hill Publishers, 1997.